

Ch-4.6 Cryptography

① Classical Cryptography

① Caesar's Cipher.

$$f(P) = (C + P) \bmod 26$$

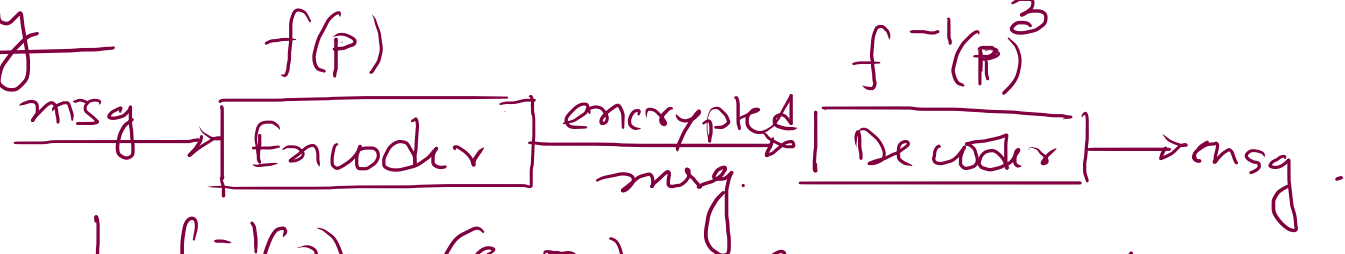
$$\% \rightarrow 0 \leq x < d \rightarrow [0, d[$$

msg = "MEET YOU"

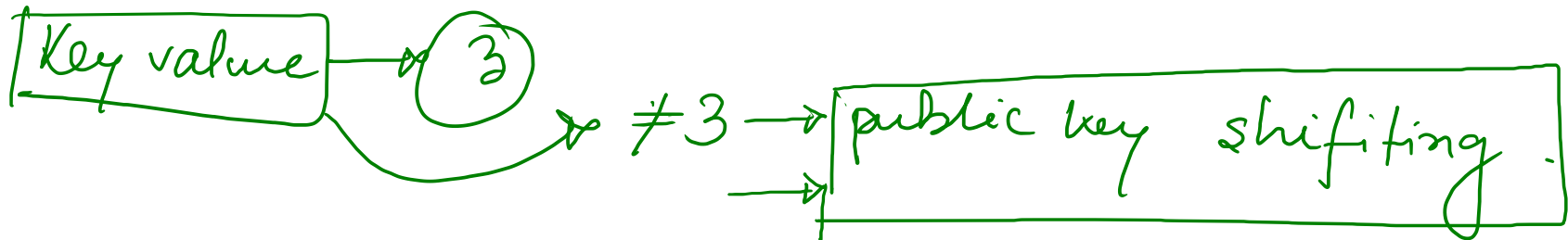
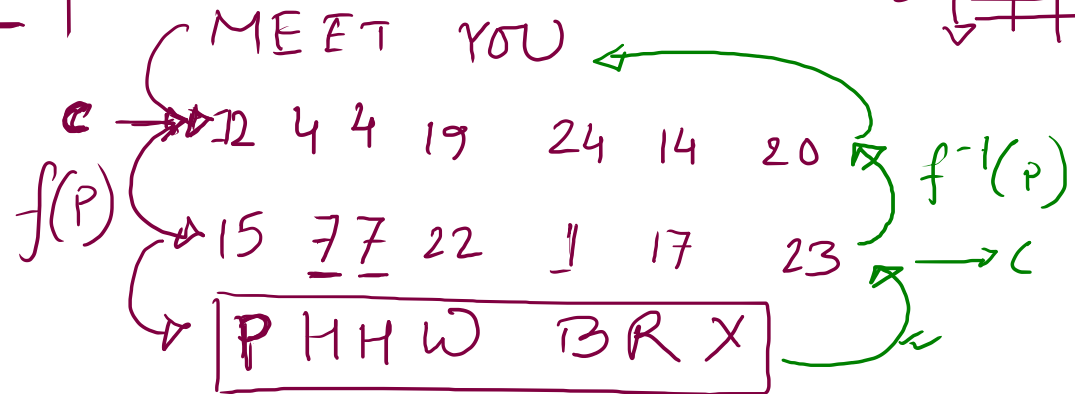
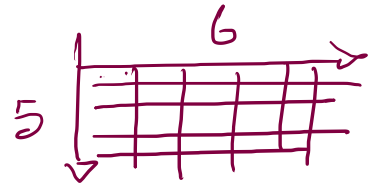
1st letter $\rightarrow A \rightarrow (1-1) = 0$

10th " $\rightarrow J \rightarrow (10-1) = 9$

13th " $\rightarrow M \rightarrow (13-1) = 12$



$$f^{-1}(P) = (C - P) \bmod 26$$



④ Block Cipher.

- Divide the letters into blocks of size m .
- if # of letters in the msg, is not divisible by m , we will add random letters at the end to make it divisible.
- A transposition function, σ is used to rearrange the alphabets.

$$\sigma(i) = m$$

→ PIRATES_x ATTACK

$$m = |\sigma| \quad \left\{ \begin{array}{l} \sigma = \{1, 2, 3, 4\} \rightarrow \sigma \\ \sigma(1) = 3 \quad \sigma(3) = 4 \\ \sigma(2) = 1 \quad \sigma(4) = 2 \end{array} \right.$$

$$\text{msg} = 13 > 4 \nmid 13 \quad 4 \mid 16$$



- PIRA TESA TIAC KAAA

$$\sigma(2) = 1$$

$$\sigma(4) = 2$$

$$\sigma(1) = 3$$

$$\sigma(3) = 4$$

APR EATS TCTA
AKA

$$\sigma^{-1}(1) = 2$$

$$\sigma^{-1}(2) = 4$$

$$\sigma^{-1}(3) = 1$$

$$\sigma^{-1}(4) = 3$$

PIRA TESA TIAC

KAAA

Wifi

→ WPA2/AES